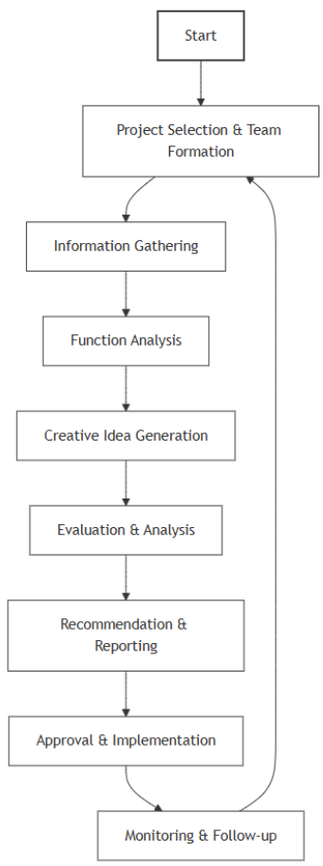
	Document No.	Section No.
	Effective Date:	Revision No.
	PROCEDURAL MANUAL	
	VALUE ENGINEERING	

OBJECTIVE	This procedure outlines the overall process for design and planning activities undertaken by BHBI. The aim is to ensure that designs are developed in a systematic and coordinated manner, meet client requirements, comply with relevant codes and standards, and are constructible, safe, and cost-effective.
SCOPE	This procedure applies to all design and planning activities performed by BHBI, from initial project conceptualization through to the completion of construction documents. It encompasses all design disciplines involved in BHBI projects (e.g., architectural, structural, MEP).

Responsibilities

- **Principal Designer:** Overall responsibility for the design and planning process. Leads the design team, coordinates with consultants, and ensures that the design meets project requirements.
- **Project Engineer:** Responsible for managing the project during the design phase, including budget, schedule, and communication with the client. They also play a key role in constructability reviews.
- **Project Team:** All members of the design team, including architects, engineers, and other specialists, are responsible for contributing to the design process and adhering to this procedure.
- **Quantity Surveyor:** Provides cost estimating and cost management input throughout the design process.
- **Procurement:** Provides input on material selection, availability, and lead times.

Flowchart



Procedure

1. Project Initiation and Brief Development:

- **Client Consultation:** The Project Engineer and Principal Designer meet with the client to understand their needs, objectives, and project requirements.
  - **Deliverables:** Client Meeting Minutes
- **Project Brief:** Develop a detailed project brief that outlines the project scope, objectives, budget, schedule, functional requirements, performance criteria, and any specific client requirements.
  - **Deliverables:** Project Brief Document
- **Project Initiation Document (PID):** If required, prepare a PID that formally authorizes the project and outlines the initial project parameters. (This may be part of a separate *Project Charter Development* procedure).
  - **Deliverables:** Project Initiation Document (PID)

2. Conceptual Design:

- **Concept Development:** The Principal Designer, in collaboration with the design team, develops initial design concepts based on the project brief.
  - **Deliverables:** Concept Design Sketches, Preliminary Drawings, Conceptual Renderings
- **Feasibility Studies:** Conduct feasibility studies to assess the viability of different design options, considering factors such as site conditions, regulatory constraints, and budget.

- **Option Analysis:** Evaluate alternative design concepts and select a preferred option in consultation with the client.
- **Preliminary Cost Estimate:** The Quantity Surveyor prepares a preliminary cost estimate based on the chosen concept design.
  - **Deliverables:** Preliminary Cost Estimate
- **Client Review and Approval:** Present the concept design and preliminary cost estimate to the client for review and approval.
  - **Deliverables:** Concept Design Report, Client Approval

3. Schematic Design:

- **Design Development:** The Principal Designer and design team refine the approved concept design, developing it into a more detailed schematic design.
  - **Deliverables:** Schematic Design Drawings, Outline Specifications
- **Coordination:** Coordinate design work between different disciplines (e.g., architectural, structural, MEP) to ensure that all systems are integrated and compatible.
- **Constructability Review:** The Project Engineer conducts constructability reviews to identify potential construction challenges and suggest design improvements.
- **Value Engineering:** Conduct a Value Engineering study, if applicable, to identify opportunities to optimize the design for cost, schedule, and performance. (Refer to the *Value Engineering Procedure*)
- **Updated Cost Estimate:** The Quantity Surveyor updates the cost estimate based on the schematic design.
  - **Deliverables:** Updated Cost Estimate
- **Client Review and Approval:** Present the schematic design and updated cost estimate to the client for review and approval.

4. Detailed Design:

- **Design Refinement:** The Principal Designer and design team develop the approved schematic design into a fully detailed design, producing detailed drawings, specifications, and other construction documents.
  - **Deliverables:** Detailed Design Drawings, Detailed Specifications
- **Technical Submittals:** Prepare and submit technical submittals to relevant authorities for review and approval (e.g., building permits). (Refer to the *Permit and Regulatory Compliance* procedure).
- **Coordination Meetings:** Hold regular coordination meetings with the design team, consultants, and potentially the client to ensure that all design elements are properly coordinated.
- **Final Cost Estimate:** The Quantity Surveyor prepares a final cost estimate based on the detailed design.
  - **Deliverables:** Final Cost Estimate
- **Client Review and Approval:** Present the detailed design and final cost estimate to the client for review and approval.

5. Construction Documentation:

- **Construction Drawings:** The Principal Designer and design team finalize the construction drawings, incorporating any approved changes and ensuring that they are complete, accurate, and coordinated.
  - **Deliverables:** Construction Drawings
- **Specifications:** Prepare detailed specifications that describe the materials, workmanship, and performance requirements for all aspects of the construction work.
  - **Deliverables:** Construction Specifications
- **Bill of Quantities (BOQ):** The Quantity Surveyor prepares a detailed BOQ based on the construction documents.
  - **Deliverables:** Bill of Quantities (BOQ)
- **Document Control:** The Office Engineer manages the distribution, revision control, and archiving of all design documents according to the *Drawing Management and Version Control Procedure*.
- **Issue for Construction:** The construction documents are formally issued for construction.

6. Design Management During Construction:

- **Respond to RFIs:** The Principal Designer and design team respond to Requests for Information (RFIs) from the construction team, providing clarifications and interpretations of the design documents. (Refer to a separate *RFI Management* procedure if BHBI develops one).
- **Review Submittals:** The Principal Designer and design team review shop drawings, material samples, and other submittals from contractors and suppliers to ensure compliance with the design intent.
- **Site Visits:** The Principal Designer and design team conduct periodic site visits during construction to observe the progress of the work and address any design-related issues that may arise.
- **Change Management:** Manage any design changes that may be required during construction, following the *Variation Order Procedure*.

7. Continuous Improvement:

- **Lessons Learned:** Conduct post-project reviews to identify lessons learned from the design and planning process.
- **Procedure Updates:** Periodically review and update this procedure to incorporate lessons learned and industry best practices.

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